

Model Selection

— Comparisons of nested linear models

STAT 24510-30040

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We have studied model comparison and model selection problem for ANOVA linear models, where nested models are compared by their sequential RSS, the residual sums of squares.

In this section we generalize this model selection procedure to general linear models, using matrix notation.

Assume the dataset consists of n observations

$$(x_{i1}, \dots, x_{i,p-1}, y_i), \quad i = 1, \dots, n.$$

We model the data by the linear regression model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0_n, \sigma^2 I_n)$$

Equivalently, the linear model can be expressed as

$$Y \sim N_n(X\beta, \sigma^2 I_n)$$

We can estimate the model parameters using the maximum likelihood method. The ML estimator $\hat{\beta}$ for the parameter vector $\beta \in \mathbb{R}^p$ agrees with the LSE, the estimator obtained via the method of least squares.

When $\text{rank}(X) = p$ with $p < n$,

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

We write the model fitted response (mean) values of the response variable as

$$\widehat{\mathbb{E}(Y)} = X\hat{\beta} = \hat{Y}$$

We are interested in comparing the linear model with smaller models containing fewer x variables than the full model.

In general, we would like to investigate methods of comparing two models which use the same set of observed data, while one is a smaller model in that it is a special case of the other larger model.

We start with comparing the smallest model, termed the null model (also called the mean model) — the model which contains the intercept term only hence no explanatory variables — with the full model containing the intercept and all of the $p - 1$ explanatory variables.

The linear regression model at the observation level can be written as

$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir} + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n.$$

The model means every explanatory variable x_i plays a role in explaining the variation in the response variable Y , thus we call it the full model.

We compare the full model with the null model

$$M_0: \quad Y_i = \beta_0 + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n.$$

The null model says that, none of the explanatory variables, i.e., none of the x_i 's, play a role in explaining the variation in the response variable Y . All variations are noise.

The comparison is equivalent to the hypothesis test

$$\begin{cases} H_o: & \beta_1 = \dots = \beta_{p-1} = 0 \\ H_a: & \text{otherwise.} \end{cases} \quad (1)$$

ANOVA identity

The full model estimated values for Y is $\hat{Y} = X\hat{\beta}$.

$$\text{The null model estimated values for } Y \in \mathbb{R}^n \text{ is } \bar{Y} = \begin{bmatrix} \bar{y} \\ \bar{y} \\ \vdots \\ \bar{y} \end{bmatrix}.$$

There is a pythagorean relationship

$$\|Y - \bar{Y}\|^2 = \|Y - \hat{Y}\|^2 + \|\hat{Y} - \bar{Y}\|^2 \quad (2)$$

In terms of projection matrix, we can rewrite (2) in the equivalent form

$$\|Y - J_n Y\|^2 = \|Y - HY\|^2 + \|HY - J_n Y\|^2 \quad (3)$$

where I_n is the identity matrix,

$$H = X(X^T X)^{-1} X^T, \quad J_n = \frac{1}{n} \mathbb{1}_n \mathbb{1}_n^T, \quad \mathbb{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

We have shown that both H and J_n are projection matrices.

To check the validity of formula (3), first, note that $\hat{Y} = X\hat{\beta} = X((X^T X)^{-1} X^T Y) = HY$ by definition. Next, check $J_n Y = \bar{Y}$. By vector inner product,

$$J_n Y = \frac{1}{n} \mathbb{1}_n \mathbb{1}_n^T Y = \frac{1}{n} \mathbb{1}_n \langle \mathbb{1}_n, Y \rangle = \frac{1}{n} \mathbb{1}_n \left(\sum_{i=1}^n y_i \right) = \frac{1}{n} \begin{bmatrix} \sum_{i=1}^n y_i \\ \vdots \\ \sum_{i=1}^n y_i \end{bmatrix} = \begin{bmatrix} \bar{y} \\ \vdots \\ \bar{y} \end{bmatrix} = \bar{Y}$$

This shows that (2) and (3) are equivalent.

In the following we prove (3).

Proof. Applying

$$\|a - b\|^2 = \|a\|^2 + \|b\|^2 - 2a^T b$$

the left hand side of (3) can be expanded into three terms.

$$\begin{aligned}\|Y - J_n Y\|^2 &= \|Y - HY + HY - J_n Y\|^2 \\ &= \|Y - HY\|^2 + \|HY - J_n Y\|^2 + 2(Y - HY)^T (HY - J_n Y)\end{aligned}$$

To show the last cross-product term is 0, we need to show $HJ_n = J_n$.

Since $H = X(X^T X)^{-1} X^T$ is a projection matrix, $H^2 = H$, and $HX = X(X^T X)^{-1} X^T = X$. That is,

$$HX = H[\mathbb{1} \ X_1 \ \cdots \ X_{p-1}] = [\mathbb{1} \ X_1 \ \cdots \ X_{p-1}]$$

where we write out the column vectors of the design matrix X . In particular, for the first column, $H\mathbb{1} = \mathbb{1}$. Thus

$$HJ_n = H\left(\frac{1}{n}\mathbb{1}\mathbb{1}^T\right) = \frac{1}{n}(H\mathbb{1})\mathbb{1}^T = \frac{1}{n}\mathbb{1}\mathbb{1}^T = J_n$$

Applying $H^2 = H$ and $HJ_n = J_n$, we obtain

$$(Y - HY)^T (HY - J_n Y) = Y^T (I_n - H)^T (H - J_n) Y = Y^T (H - J_n - H^2 + HJ_n) Y = 0$$

Therefore,

$$\|Y - J_n Y\|^2 = \|Y - HY\|^2 + \|HY - J_n Y\|^2$$

which is equivalent to

$$\|Y - \bar{Y}\|^2 = \|Y - \hat{Y}\|^2 + \|\hat{Y} - \bar{Y}\|^2$$

We have proved both (2) and (3). □

Relation with the SS decomposition in ANOVA

If we write

$$\|Y - \bar{Y}\|^2 = \|Y - \hat{Y}\|^2 + \|\hat{Y} - \bar{Y}\|^2$$

in terms of the components of the vectors, we obtain our familiar sums of squares decomposition in ANOVA,

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

In SS notation for the sum of squares,

$$SS_{total} = SS_e + SS_{model}$$

In RSS notation for the residual or the error sum of squares, since the fitted null model is the mean (thus also called the mean model), other common notations are

$$SS_{total} = RSS_{null}, \quad SS_e = SS_{error} = RSS_{full} = RSS.$$

Properties of the SS decomposition (Theorem)

The following properties are essential in the hypothesis testing problem (1) and similar tests on nested models.

(a) $\frac{SS_e}{\sigma^2} \sim \chi_{n-p}^2$.

The proof is analogous to the one for ANOVA nested model comparisons.

(b) $SS_e \perp\!\!\!\perp SS_{model}$.

That is, SS_e and SS_{model} are independent.

This important fact can be used to compare the two SS 's.

(c) Under the null

$$H_o : \beta_1 = \cdots = \beta_{p-1} = 0$$

(i) $\frac{SS_{model}}{\sigma^2} \sim \chi_{p-1}^2$

(ii) $\frac{SS_{total}}{\sigma^2} \sim \chi_{n-1}^2$.

(iii) $\frac{SS_{model}/(p-1)}{SS_e/(n-p)} \sim F_{p-1, n-p}$.

Applications in model comparison

- Based on the above properties, to test the hypothesis (1),

$$\begin{cases} H_o : & \beta_1 = \cdots = \beta_{p-1} = 0 \\ H_a : & \text{otherwise.} \end{cases}$$

we can use the F test

$$F = \frac{SS_{model}/(p-1)}{SS_e/(n-p)} = \frac{(RSS_{null} - RSS_{full})/((n-1) - (n-p))}{RSS_{full}/(n-p)} \sim F_{p-1, n-p} \quad \text{under } H_o$$

and reject the null when the ratio is too large, say, when $F > F_{p-1, n-p, 1-\alpha}$.

Reject the null means rejecting the null model (the mean-only model), meaning the data show that at least some explanatory variable x_i 's play a significant role in explaining the variations in the response variable y .

- Analogously, similar properties hold for two nested models, where the smaller model is a special case of the larger model.

F statistic can be constructed to compare any two nested models, e.g., testing

$$\begin{cases} H_o^* : & \beta_{q+1} = \cdots = \beta_{p-1} = 0, \quad q < p, \\ H_a : & \text{all } \beta_i \neq 0, \quad i = 0, 1, \dots, p-1. \end{cases}$$

The corresponding F test is

$$\frac{(RSS_{small} - RSS_{large})/((n-q) - (n-p))}{RSS_{large}/(n-p)} \sim F_{p-q, n-p} \quad \text{under } H_o^*$$

just as we have claimed in the comparison of nested models for ANOVA models.

Proofs of the properties of the SS decomposition

Some immediate results

- Note that (ii) is an immediate consequence of properties (a), (b), (c-i) and the definition of χ^2 distribution. because

$$SS_{total} = SS_e + SS_{model}$$

- Property (c-ii)

$$\frac{SS_{total}}{\sigma^2} \sim \chi_{n-1}^2$$

agrees to the one-sample case of sample size n , where we have derived that, for sample variance S^2 ,

$$(n-1)S^2 \sim \chi_{n-1}^2$$

under the normal assumption.

- The last property (c-iii) follows from the definition of F distribution.

Proof of (a) — The distribution of SS_e

Proof. To show (a), note that the residual vector then can be written as

$$\begin{aligned} \hat{\varepsilon} &= Y - \hat{Y} = (X\beta + \varepsilon) - X\hat{\beta} \\ &= (X\beta + \varepsilon) - X(X^T X)^{-1} X^T Y \\ &= (X\beta + \varepsilon) - X(X^T X)^{-1} X^T (X\beta + \varepsilon) \\ &= X\beta + \varepsilon - X\beta - X(X^T X)^{-1} X^T \varepsilon \\ &= \varepsilon - H\varepsilon = (I_n - H)\varepsilon \end{aligned}$$

Then

$$SS_e = RSS = \sum_{i=1}^n \hat{\varepsilon}_i^2 = ((I - H)\varepsilon)^T ((I - H)\varepsilon) = \varepsilon^T (I - H)^T (I - H)\varepsilon = \varepsilon^T (I - H)\varepsilon$$

where we have used the symmetry and projection property of H ,

$$H^T = H, \quad H^2 = H$$

Since H is a projection matrix, $I_n - H$ is also a projection matrix. Use the matrix trace property,

$$\text{rank}(I_n - H) = \text{trace}(I_n - H) = \text{trace}(I_n) - \text{trace}(H) = n - p$$

Write

$$\varepsilon = \sigma Z$$

then

$$Z = \frac{1}{\sigma} \varepsilon \sim N_n(0, I_n)$$

Therefore

$$\frac{SS_e}{\sigma^2} = \frac{\varepsilon^T (I - H)\varepsilon}{\sigma^2} = Z^T (I - H)Z$$

where $I - H$ is a projection matrix of rank $n - p$, and $Z \sim N_n(0, I_n)$. By the result we obtained from projection matrix,

$$\frac{SS_e}{\sigma^2} \sim \chi_{n-p}^2$$

□

Proof of (b) — The independence of SS_e and SS_{model}

Proof. To show (b) that

$$SS_e \perp\!\!\!\perp SS_{model}$$

let's use the vector-matrix expression in (3),

$$\begin{aligned} SS_e &= \|Y - HY\|^2 = \|(I_n - H)Y\|^2, \\ SS_{model} &= \|HY - J_n Y\|^2 = \|(H - J_n)Y\|^2. \end{aligned}$$

To show the independence relation $SS_e \perp\!\!\!\perp SS_{model}$, it is sufficient to show $(I_n - H)Y \perp\!\!\!\perp (H - J_n)Y$.

Both $(I_n - H)Y$ and $(H - J_n)Y$ are linear transformations of $Y \sim N_n(X\beta, \sigma^2 I_n)$.

Consequently, $(I_n - H)Y$ and $(H - J_n)Y$ are jointly normal (analogous to the reasoning for bivariate normal).

Therefore, to show $(I_n - H)Y \perp\!\!\!\perp (H - J_n)Y$, it is sufficient to show that they are uncorrelated.

$$\begin{aligned} \text{Cov}((I_n - H)Y, (H - J_n)Y) &= (I_n - H) \text{Cov}(Y, Y) (H - J_n)^T \\ &= (I_n - H) \sigma^2 I_n (H - J_n) \\ &= \sigma^2 (I_n - H)(H - J_n) \\ &= \sigma^2 (H - J_n - H^2 + HJ_n) \end{aligned}$$

We have already shown that

$$HJ_n = J_n$$

Therefore,

$$\text{Cov}((I_n - H)Y, (H - J_n)Y) = \sigma^2 (H - J_n - H^2 + HJ_n) = \sigma^2 (H - J_n - H + J_n) = 0_{n \times n}$$

The normality means the uncorrelated-ness implies independence, so

$$(I_n - H)Y \perp\!\!\!\perp (H - J_n)Y \quad \Rightarrow \quad SS_e \perp\!\!\!\perp SS_{model}$$

□

Proof of (c) — The distribution of SS_{model} under H_o

Proof. To show (c), that under the null,

$$\frac{SS_{model}}{\sigma^2} = \frac{\|(H - J_n)Y\|^2}{\sigma^2} \sim \chi_{p-1}^2,$$

recall that the H_0 null model in (1) is

$$Y_i = \beta_0 + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n.$$

which can be written as

$$Y \sim N_n(\beta_0 \mathbb{1}_n, \sigma^2 I_n)$$

So we may write

$$Y = \beta_0 \mathbb{1}_n + \varepsilon = \beta_0 \mathbb{1} + \sigma Z,$$

where

$$Z = \frac{1}{\sigma} \varepsilon \sim N_n(0, I_n).$$

Then

$$\begin{aligned} (H - J_n)Y &= (H - J_n)(\beta_0 \mathbb{1}_n + \sigma Z) \\ &= (H - J_n)\beta_0 \mathbb{1}_n + (H - J_n)\sigma Z \\ &= \beta_0(H - J_n)\mathbb{1}_n + \sigma(H - J_n)Z \end{aligned}$$

Since $H\mathbb{1}_n = \mathbb{1}_n$ as proved previously,

$$(H - J_n)\mathbb{1}_n = H\mathbb{1}_n - \left(\frac{1}{n}\mathbb{1}_n\mathbb{1}_n^T\right)\mathbb{1}_n = H\mathbb{1}_n - \frac{1}{n}\mathbb{1}_n(\mathbb{1}_n^T\mathbb{1}_n) = \mathbb{1}_n - \left(\frac{1}{n}\mathbb{1}_n\right)n = \mathbb{1}_n - \mathbb{1}_n = 0_n$$

Thus

$$(H - J_n)Y = \sigma(H - J_n)Z$$

Now we need to show that $H - J_n$ is a projection matrix.

The matrix $H - J_n$ is symmetric,

$$(H - J_n)^T = H - J_n$$

Note that

$$(HJ_n)^T = J_nH = (HJ_n)^T = J_n^T = J_n.$$

Then we can derive the idempotent property,

$$\begin{aligned} (H - J_n)^2 &= (H - J_n)(H - J_n) \\ &= H^2 - HJ_n - J_nH + J_nJ_n = H - J_n - J_n + J_n = H - J_n \end{aligned}$$

Hence the matrix $H - J_n$ is a projection matrix, its rank is

$$\text{rank}(H - J_n) = \text{trace}(H - J_n) = \text{trace}(H) - \text{trace}(J_n) = p - 1$$

Therefore, since $H - J_n$ is a projection matrix of rank $p - 1$ and

$$Z \sim N_n(0, I_n)$$

by the property of projection matrix,

$$\|(H - J_n)Z\|^2 \sim \chi_{p-1}^2$$

Now we have proved that, under the null,

$$\frac{SS_{model}}{\sigma^2} = \frac{\|(H - J_n)Y\|^2}{\sigma^2} = \|(H - J_n)Z\|^2 \sim \chi_{p-1}^2$$

□

Generalization of the ANOVA identity

Recall that, the estimator $\hat{\beta}$ minimizes the residual sums of squares (RSS) of the model,

$$RSS = \sum_{i=1}^n \left(Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_2 x_{i,r}) \right)^2 = \|Y - X\hat{\beta}\|^2$$

In other words, $\|Y - X\hat{\beta}\|^2$ is the smallest among all $\|Y - X\beta\|^2$ for any $\beta \in \mathbb{R}^p$.

$$\|Y - X\hat{\beta}\|^2 \leq \|Y - X\beta\|^2, \quad \forall \beta \in \mathbb{R}^p.$$

In fact, there is a general pythagorean relationship

$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\beta\|^2, \quad \forall \beta \in \mathbb{R}^p. \quad (4)$$

(You may have recognized that this as an exercise in a homework assignment.)

In particular, we may choose

$$\beta = \hat{\beta}_{null} = \begin{bmatrix} \hat{\beta}_{0,null} \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^p = \begin{bmatrix} \frac{1}{n}\mathbb{1}^T Y \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^p \quad \Rightarrow \quad X\beta = X\hat{\beta}_{null} = \begin{bmatrix} \bar{y} \\ \bar{y} \\ \vdots \\ \bar{y} \end{bmatrix} = \bar{Y} \in \mathbb{R}^n$$

Then (4) becomes

$$\|Y - \bar{Y}\|^2 = \|Y - \hat{Y}\|^2 + \|\hat{Y} - \bar{Y}\|^2$$

That is equation (2) we have proved.